Title:	**Access Application Development: Reports and Forms**
Special Notes / Additional Resources:	• This assignment is completed using a single database. As a precaution, make backup copies of your database as you complete significant portions of the work in case of accidental error or deletion. • Hand in your single, completed database.

Part A: Report Design (5 marks)

Setup

1. Download the Starter Database and save and rename it to include your name and student number (e.g., Sousa-12345-AS5.accdb).

Instructions

1. **[2 marks]** Create a summary query called <yourLastName>SalesRepDiscountSummary (e.g., SousaSalesRepDiscountSummary) that includes the following columns.
   - Sales Rep Number
   - Sales Rep First Name
   - Sales Rep Last Name
   - Sum of the revenue (a summation of the qty ordered * quoted price)
   - Sum of the potential revenue (a summation of the qty ordered * unit price from the parts table)
   - Sum of the discount which is a sum of (qty ordered * unit price from the parts table) - (qty ordered * quoted price)

   Group the summary query by Sales Rep Number, First name, and Last name

   Ensure meaningful names are used as the column headings in the query results.

   When correctly formed, your query should appear as follows (the data values may be different in your version of the database). Notice that meaningful column headings have been specified for the summary items.

SousaSalesRepDiscountSummary

SalesRepNu	LAST	FIRST	Revenue	PotentialRe	Discount
03	Jones	Mary	25760.22	40191.22	14431
06	Smith	William	1282.41	1384.51	102.1
12	Diaz	Miguel	3487.58	3859.5	371.92

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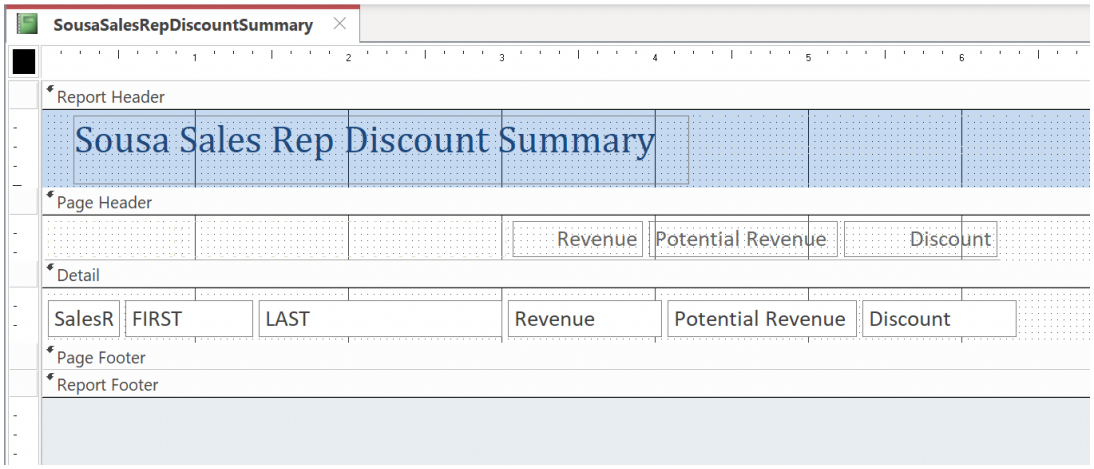
2. **[3 marks]** Create a Report design, which uses the query above as its data source. Save the report as <yourLastname>SalesRepDiscountSummary (e.g.,SousaSalesRepDiscountSummary") and layout the report as follows:

   Layout the design view of the report as follows:
   a. Add a report title (label) with your name as the prefix.
   b. Set the layout of the fields on the report to use a tabular arrangement (tip, add all the fields first, then select all the fields and choose "arrange, tabular" from the ribbon).
   c. Remove the field headings for the sales rep number, first name, and last name.
   d. Resize the fields so that the data fits on an 8.5x11 page in portrait layout and that all the data is visible.
   e. Remove the default border that appears on all the objects on the report (select all the elements on the report and set the border to transparent using the format ribbon).
   f. Set the format of the dollar amounts on the report so that they appear as currency.



3. Once complete, the report should appear to the example below. Be sure to save the final report/changes.

Part B: Form Design (17 Marks)

Overview

This section of the assignment evaluates your ability to design an effective and well laid out form to be used by a non-technical user. You will be evaluated on both the functionality and the aesthetics of the form design (you can have a bit of fun with it as well).

Focus/Purpose of the Form: Part Entry/Maintenance

Allows the user to maintain data held in the "Part" table. To help the user evaluate what unit price to establish or revise, the form also provides a historical sales summary of that part.

Getting Started / Before you Start Form Design

1. Before starting your form design, create the query that will be used to summarize the sales of parts grouped by sales year and part number. The query is to be sorted descending by year (most recent year at the top). Its output, when created correctly, should appear similar to the example below (values in your version of the database may differ):

Year	PART_NUMBER	Qty Sold	Average Quoted Price	Revenue
2007	CB03	25	$21.34	$533.50
2006	CX11	100	$11.25	$1,125.00
2005	AX12	25	$20.56	$514.00
2005	CB03	10	$269.99	$2,699.90
2004	AX12	100	$18.00	$1,800.00
2004	AZ52	50	$12.87	$643.50
2003	AZ52	50	$12.87	$643.50
2002	AX12	11	$21.95	$241.45
2002	AZ52	211	$11.62	$1,932.45
2002	BA74	4	$24.95	$99.80
2002	BT04	2	$149.99	$299.98
2002	BZ66	1	$399.99	$399.99
2002	CB03	74	$264.99	$18,899.26
2002	CX11	2	$22.95	$45.90
2002	CZ81	2	$325.99	$651.98

Year: The year based on the order date (sorted descending)

Quantity Sold: the sum of the quantity sold on the order detail

Average Quoted Unit Price: the average quoted price from the order detail

Revenue: A sum of the quoted price * quantity sold

Save this query as <yourName>ProductSalesHistory (e.g., "SousaProductSalesHistory")

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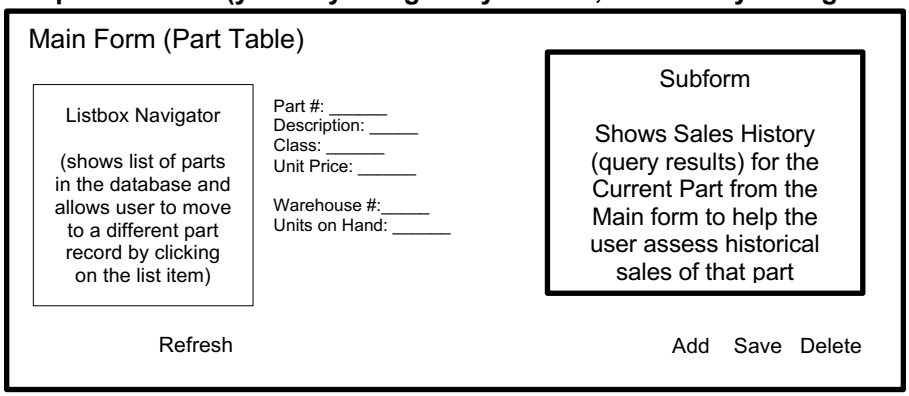
Overview
This design will make use of a main form and a subform. The main form allows for editing and management of data in the parts table. The subform is used to show the user historical sales for the part they are currently editing in the main form.

Concept Wireframe (you may design as you wish, this is only a rough idea)



Form Layout/Design Aesthetics and Validation Technique [5 Marks]
Attention to logical layout, proper labels, alignment, size, and placement of buttons...etc. will be evaluated. The means of validation used (preventing errors when possible) will also be evaluated.

Form Features that Must Be Implemented [12 marks]

1. **[1 mark]** The Form must be a pop-up window and appropriately sized.

2. **[4 marks]** The form must allow editing of the part table (all fields) and should implement the following data validation rules as follows:
    a. Warehouse Number can only be a value of: 1, 2, or 3
    b. Item class must be one of: HW, SG, or AP
    c. The part number must be in the format of AA99 (two letters followed by two numbers)
    d. The part Unit price must be greater than zero (no negative values allowed)

3. **[2 marks]** A list box is to be used as a navigational control to allow the user to navigate from one part to another. The list box should display the part description.

4. **[2 marks]** The form must provide the following buttons/actions:
    a. Add (adds a new part)
    b. Save (saves changes to the part)
    c. Delete (deletes a part)
    d. Close (closes the form)
    e. Refreshes (refresh the form)

5. **[3 marks]** In a subform, a summary of the parts' sales history by year will appear based on the query created earlier in the assignment.. As the user moves from one part to another on the main form, the subform will change to the sales history of the part.

    The layout and aesthetics of this subform should provide an easy to review set of statistics that excludes the part number which would be redundant as it already appears in the parent form.